

Cloud Computing

Chapter 5 – SLA

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What is SLA?

Definition:

A Service Level Agreement (SLA) is the bond for performance negotiated between the cloud services provider and the client.

- A **Service Level Agreement (SLA)** is a contract between:
 - Cloud Service Provider
 - Client (Consumer)
- Defines expected level of service performance
- Modern cloud providers mostly offer **standardized SLAs**

Types of SLA

- **Customer-based SLA**
- **Service-based SLA**
- **Multilevel SLA**

These SLAs define how services are structured and delivered to customers.

Customer-based SLA

Definition:

- SLA tailored for a **specific customer or organization**

Key Features:

- Customized agreement
- Covers all services used by that customer
- Focuses on customer-specific requirements

Example:

- A bank signs an SLA with a cloud provider for:
 - 99.99% uptime
 - High security
 - 24/7 support

Advantage: Highly flexible and customer-focused

Disadvantage: Complex to manage

Service-based SLA

Definition:

- SLA defined for a **specific service** and applies to all customers

Key Features:

- Same SLA for all users
- Standardized and easy to manage
- Common in public cloud services

Example:

- Cloud storage service offering:
 - 99.9% uptime
 - Fixed response time
- Applies to all customers equally

Advantage: Simple and cost-effective

Disadvantage: Less flexibility for specific needs

Multilevel SLA

Definition:

- SLA structured into **multiple levels** to address different needs

Levels:

- **Corporate Level:** Applies to entire organization
- **Customer Level:** Specific to a group of customers
- **Service Level:** Specific to a particular service

Example:

- Company-wide security policy (corporate)
- Department-specific uptime needs (customer)
- Email service SLA (service level)

Advantage: Flexible and scalable

Disadvantage: More complex to design and manage

Comparison of SLA Types

Feature	Customer	Service	Multilevel
Customization	High	Low	Medium
Complexity	High	Low	High
Flexibility	High	Low	High
Usage	Enterprises	Public Cloud	Large Organizations

Key SLA Parameters

- **Availability (Uptime):** Percentage of time the service is operational and accessible.
- **Latency / Response Time:** Time taken by the system to respond to a user request.
- **Reliability of Service Components:** Ability of system components to function without failure over time.
- **Accountability of Each Party:** Clearly defined responsibilities of both provider and customer.
- **Warranties:** Assurances or guarantees provided regarding service quality and performance.

Uptime vs Downtime

Uptime:

- Percentage of time a system/service is **operational and available**
- Example: 99.9% uptime means service is available most of the time

Downtime:

- Percentage of time a system/service is **not available or fails**
- Includes failures, maintenance, or outages

Relation:

- $\text{Uptime} + \text{Downtime} = 100\%$
- Higher uptime $\Rightarrow$ Lower downtime

Example:

- 99.9% uptime $\Rightarrow$ 0.1% downtime $\approx$ 8.76 hours/year

Example: Windows Azure SLA & SQL Azure SLA

Below are two major Service Level Agreements (SLA) described:

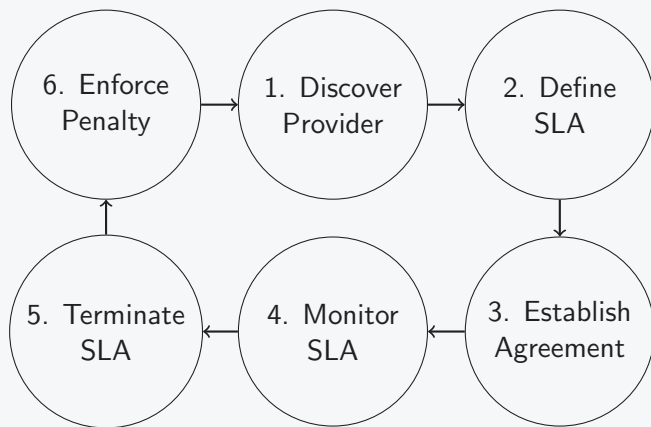
Windows Azure SLA:

- Separate SLAs for **Compute** and **Storage**
- **99.95% uptime** when deployed across fault and upgrade domains
- **99.9% detection guarantee** for role instance failures

SQL Azure SLA:

- Ensures connectivity between database and gateway
- **99.9% monthly availability**
- Availability = (Available time / Total time in month)
- Measured over a **30-day cycle**
- Downtime counted when connection requests are denied

SLA Lifecycle



SLA Lifecycle

- ➊ **Discover Service Provider:** Identify and evaluate a provider that meets business and technical requirements.
- ➋ **Define SLA:** Specify service expectations such as uptime, performance, security, and response time.
- ➌ **Establish Agreement:** Formally document terms, conditions, responsibilities, and penalties agreed by both parties.
- ➍ **Monitor SLA Violations:** Continuously track service performance to ensure compliance with agreed SLA metrics.
- ➎ **Terminate SLA (if required):** End the agreement if service levels are not met or business needs change.
- ➏ **Enforce Penalties:** Apply compensation or penalties if the provider fails to meet SLA commitments.

E-commerce company :SLA Lifecycle

Scenario: An e-commerce company (like Amazon/Flipkart) uses cloud services.

- ➊ **Discover Service Provider:** Company selects a cloud provider (e.g., AWS/Azure) based on cost, reliability, and scalability.
- ➋ **Define SLA:** Requirements are defined such as 99.9% uptime, fast response time, and high security.
- ➌ **Establish Agreement:** A formal SLA is signed including service terms, responsibilities, and penalties.
- ➍ **Monitor SLA Violations:** Performance is continuously monitored using tools to check uptime and response time.
- ➎ **Enforce Penalties:** If downtime occurs during a sale, the provider compensates as per SLA.
- ➏ **Terminate SLA:** If issues persist, the company may switch to another provider.

SLA Lifecycle : Online Banking

Scenario: A bank uses cloud services for online banking operations.

- ➊ **Discover Service Provider:** Bank selects a secure and reliable cloud provider with strong compliance support.
- ➋ **Define SLA:** Bank requires 99.99% uptime, high security, and quick transaction processing.
- ➌ **Establish Agreement:** SLA is signed including data protection, uptime guarantees, and penalty clauses.
- ➍ **Monitor SLA Violations:** Continuous monitoring ensures transactions are processed without delay or failure.
- ➎ **Enforce Penalties:** If system downtime affects transactions, the provider compensates the bank.
- ➏ **Terminate SLA:** If security or performance issues persist, the bank switches providers.

SLA Lifecycle: Video Streaming Platform

Scenario: A global video streaming platform (like Netflix) uses cloud services to stream content worldwide.

- ➊ **Discover Service Provider:** Platform selects a cloud provider with global data centers, CDN support, and high scalability.
- ➋ **Define SLA:** Requires 99.99% uptime, low latency streaming, multi-region availability, and auto-scaling during peak traffic.
- ➌ **Establish Agreement:** SLA includes uptime guarantees, latency limits, data redundancy, disaster recovery, and penalty clauses for outages.
- ➍ **Monitor SLA Violations:** Real-time monitoring tracks buffering, server load, and regional outages using analytics tools.
- ➎ **Enforce Penalties:** If downtime occurs during a live event, provider compensates via service credits and financial penalties.
- ➏ **Terminate SLA:** If repeated SLA failures occur, platform may adopt a multi-cloud strategy or switch providers.

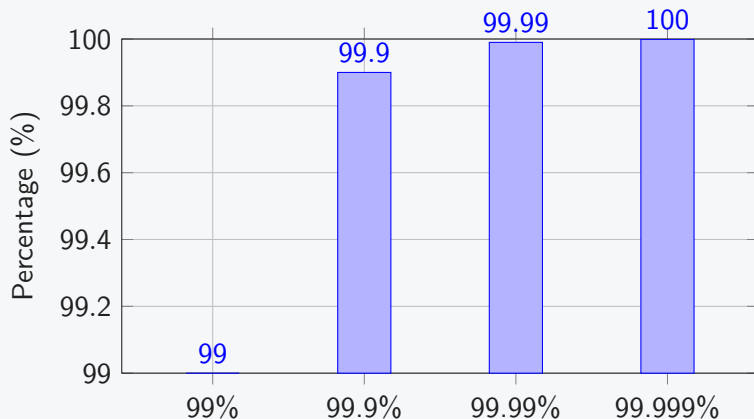
Advantages of SLA

- Improved communication
- Increased accountability
- Alignment with business goals
- Reduced downtime
- Better cost management

Disadvantages of SLA

- Complex to create and maintain
- Can be rigid
- Limited service flexibility
- Misaligned incentives
- Limited legal liability

Uptime vs Downtime Comparison



Higher SLA $\Rightarrow$ Lower Downtime

Understanding Downtime

Uptime	Downtime per Year
99%	3.65 days
99.9%	8.76 hours
99.99%	52.6 minutes
99.999%	5.26 minutes

Numerical: SLA in Streaming Platform

Scenario: A global video streaming platform guarantees **99.99% uptime**. During a major live event, the service was down for **1 hour** in a year.

Questions:

- Calculate the **maximum allowed downtime** as per SLA.
- Determine whether the **SLA is violated**.

Numerical: SLA in Streaming Platform

Scenario: A global video streaming platform guarantees **99.99% uptime**. During a major live event, the service was down for **1 hour** in a year.

Questions:

- Calculate the **maximum allowed downtime** as per SLA.
- Determine whether the **SLA is violated**.

Solution:

- Total time in a year = 8760 hours
- Allowed downtime = 0.01% of 8760

$$= 0.0001 \times 8760 = 0.876 \text{ hours}$$

Conclusion:

- Actual downtime = 1 hour
- Allowed downtime = 0.876 hours
- $\Rightarrow$ **SLA Violated**

Numerical: SLA with Revenue Impact

Scenario: A video streaming platform earns Rs.5,00,000 per hour during live events. SLA guarantees **99.9% uptime**. During a major event, service was down for **3 hours** in a month.

Questions:

- Calculate the **allowed downtime** per month (30 days).
- Check whether the **SLA is violated**.
- Calculate the **revenue loss**.

Numerical: SLA with Revenue Impact

Scenario: A video streaming platform earns Rs.5,00,000 per hour during live events. SLA guarantees **99.9% uptime**. During a major event, service was down for **3 hours** in a month.

Questions:

- Calculate the **allowed downtime** per month (30 days).
- Check whether the **SLA is violated**.
- Calculate the **revenue loss**.

Solution:

- Total time/month = $30 \times 24 = 720$ hours
- Allowed downtime = 0.1% of 720
$$= 0.001 \times 720 = 0.72 \text{ hours}$$
- Actual downtime = 3 hours
- Since $3 > 0.72 \Rightarrow$ **SLA Violated**
- Revenue loss = $3 \times \text{Rs.5,00,000} = \text{Rs.15,00,000}$

Numerical 1

Q1. A cloud service guarantees 99% uptime. Calculate yearly downtime.

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Answer:

- Total hours/year = 8760
- Downtime = 1% of 8760 = 87.6 hours
- ≈ 3.65 days

Numerical 2

Q2. If uptime is 99.9%, find downtime per year.

Numerical 2

Q2. If uptime is 99.9%, find downtime per year.

Answer:

- Downtime = 0.1% of 8760
- = 8.76 hours

Numerical 3

Q3. A service offers 99.99% uptime. Calculate downtime per month (30 days).

Numerical 3

Q3. A service offers 99.99% uptime. Calculate downtime per month (30 days).

Answer:

- Total minutes/month = 43200
- Downtime = $0.01\% = 4.32$ minutes

Numerical 4

Q4. Calculate downtime for 99.999% uptime per year.

Numerical 4

Q4. Calculate downtime for 99.999% uptime per year.

Answer:

- Downtime = 0.001% of 8760
- = 5.26 minutes

Numerical 5

Q5. A system runs for 720 hours/month but fails for 2 hours. Find availability.

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Q5. A system runs for 720 hours/month but fails for 2 hours. Find availability.

Formula:

$$\text{Availability} = \frac{\text{Total Time} - \text{Downtime}}{\text{Total Time}} \times 100$$

Answer:

$$= \frac{720 - 2}{720} \times 100 = 99.72\%$$

Numerical 6

Q6. SLA guarantees 99.9% uptime, actual uptime is 99.7%. Is SLA violated?

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Answer:

- Yes, actual uptime $<$ SLA
- SLA is violated

Numerical 7

Q7. For 99% uptime, calculate downtime per day.

Numerical 7

Q7. For 99% uptime, calculate downtime per day.

Answer:

- Total minutes/day = 1440
- Downtime = 1% = 14.4 minutes/day

Numerical 8

Q8. Two independent servers each have 99% uptime. Find combined availability.

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Q8. Two independent servers each have 99% uptime. Find combined availability.

Formula:

$$A = 1 - (1 - 0.99)^2$$

Answer:

$$= 1 - (0.01)^2 = 0.9999 = 99.99\%$$

Numerical 9

Q9. Provider offers *Rs.10,000* credit if SLA is violated. Actual uptime is 99.5% vs SLA 99.9%.

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Answer:

- SLA violated
- Penalty = *Rs.10,000*

Numerical 10

Q10. For 99.9% SLA, calculate maximum downtime per month (30 days).

Numerical 10

Q10. For 99.9% SLA, calculate maximum downtime per month (30 days).

Answer:

- Total minutes = 43200
- Downtime = $0.1\% = 43.2$ minutes

Q1

Scenario: A startup uses a cloud service with 99.9% uptime. Downtime in a month = 2 hours.

Q1

Scenario: A startup uses a cloud service with 99.9% uptime. Downtime in a month = 2 hours.

Answer:

- Total time = 720 hours
- Uptime = 718 hours
- Availability = $\frac{718}{720} \times 100 = 99.72\%$
- SLA violated

Scenario: E-commerce site faces downtime during peak sale.

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Answer:

- Affected: Availability, Performance
- Solutions:
 - Auto-scaling
 - Load balancing

Scenario: Company selects cheaper plan with low SLA.

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Answer:

- Risks: High downtime, poor reliability
- Impact: Customer loss, business disruption

Scenario: Response time promised = 200 ms, actual = 500 ms.

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Answer:

- Violated: Latency SLA
- Action: Report violation, claim compensation

Scenario: Uptime met but issue resolution is slow.

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Answer:

- Weak area: Support SLA
- Improve: Faster response time, better monitoring

Scenario: Banking app with 99.99% SLA, downtime = 2 hours/year.

Scenario: Banking app with 99.99% SLA, downtime = 2 hours/year.

Answer:

- Allowed downtime = 0.876 hours
- Actual downtime = 2 hours → SLA violated
- Improve: Redundancy, multi-region deployment
- Impact: Financial loss, customer dissatisfaction

Scenario: Provider A (99.9%), Provider B (99.99%)

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Answer:

- 99.9% → 8.76 hrs downtime
- 99.99% → 52.6 min downtime
- Critical apps → Provider B
- Trade-off: Cost vs Availability

Scenario: SLA signed but not monitored.

Scenario: SLA signed but not monitored.

Answer:

- Failed stage: Monitoring
- Use tools: CloudWatch, Azure Monitor
- Actions: Alerts, audits

Scenario: SLA violations but penalties are low.

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Answer:

- Issue: Weak penalty enforcement
- Improve: Increase penalties, stricter SLA
- Ensures accountability

Scenario: Streaming service crashes during event.

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Answer:

- Affected: Availability, Performance
- Impact: Revenue loss
- Improve: Auto-scaling, CDN
- Enhance SLA: Higher uptime guarantee